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EDUCATION

Rutgers University–New Brunswick, School of Engineering | B.S. Mechanical Engineering Class of 2030

HONORS

Sir Isaac Newton Physics Exam: 9th internationally of 3,800+ (2025) | **USAPhO:** Distinction, top 400 of 6,000 (2025)
NASA App Development Challenge: National Winner, 200+ teams (2025) | **MIT Beaver Works:** 4th of 100 teams (2025)
NJ State Science Fair: 2nd Place, Computer Science, flood model (2026); Air Force Award, drone-swarm RL (2025)

PUBLICATION

Beyond the Haystack: Sensitivity to Context in Legal Reference Recall. Third author. NLLP Workshop at EMNLP 2025.
• Evaluation framework for long-context LLM recall on three naturalistic legal-reference tasks built from U.S. court-opinion briefings; exposed behavior missed by synthetic benchmarks such as Needle in a Haystack.

RESEARCH & ENGINEERING PROJECTS

Physics-Informed Flood Forecasting Model *Science fair research* 2025–2026
• Built a graph transformer in PyTorch for flood forecasting, with the 1D Saint-Venant continuity equation in the training loss to penalize predictions that violate conservation of mass.
• In project benchmarks against a finite-difference reference, reached 117× faster inference with 97% mass-conservation adherence.

Cable-Driven Soft Robotic Arm *Independent; built and bench-tested* 2024–2026
• Built a continuum arm from segmented TPU driven by three cables; differential cable tension and material compliance produce 3D bending.
• Ran bench experiments on bending, compliance, and response; built a camera-based hand-pose control mode and a target-trajectory visualizer, and am developing a VLM + heuristic solver controller.

Kestrel Mini R4 Bicopter *Build in progress; not yet flown* 2026
• Designed a compact bicopter in FreeCAD with two lift motors and two independent tilt servos, 40 printed parts, and a 42-route wire schedule; now building the first physical prototype.
• Estimated 665 g with a calculated 1.82 thrust-to-weight at 12 V, assuming 15% installation loss. Thrust, servo tracking, print strength, and flight tests are still to come.

SA-2 Samson Tilt-Nacelle VTOL *CAD prototype; build planned after Kestrel* 2026
• Modeled a 1:15 scale RC tilt-nacelle VTOL in FreeCAD (twin tilting nacelles, coaxial 12 in rotors, ArduPilot QuadPlane), plus a two-motor printed-structure bicopter variant with documented flight-controller wiring.
• Wrote a preliminary design review that gates free flight behind staged structural, propulsion, tilt, and hover tests; tracked mass and geometry in a controlled parameter file with Python consistency checks.

Axiom: Offline Math Notebook *macOS desktop application* 2026
• Built a local notebook combining typeset equations, matrices, interactive 2D/3D graphs, multivariable calculus, a reference-PDF pane, and PDF export; notebooks stay on the computer.
• Wrote a 2D/3D truss solver using the method of joints: checks static determinacy (unknowns vs. joint equations), solves member forces and support reactions, and grades user-entered answers. Code signing and notarization in progress.

Active Aerodynamics RC Testbed *Independent; in development* 2023–2026
• Built a 1/8-scale RC car with a ground-effect downforce fan and drag-reduction system; ECU runs on a repurposed smartphone with on-device perception.

NASA App Development Challenge, Team Luna Vista *National Winner* 2025
• Modeled the Artemis II flight path in Unreal Engine 5 using orbital mechanics and C++ shaders; optimized antenna scheduling for the visualization.

SKILLS & TRAINING

CAD & simulation: SolidWorks, Fusion 360, FreeCAD, build123d, OpenFOAM (CFD), CalculiX (FEA)
Hardware: 3D-printed TPU and structural parts, cable actuation, RC electronics and wiring layouts
Software & modeling: Python, C++, Unreal Engine 5, physics-informed ML, reinforcement learning, model evaluation
Safety: OSHA-10